

Beyond Spatial Privacy: Protecting Trajectories With Spatio-Temporal Differential Privacy

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Abstract—Spatio-temporal trajectories carry identifying information and are vulnerable to privacy breaches. Existing studies predominantly focus on the spatial domain. The temporal aspect remains underexplored, leaving privacy risks unaddressed. This paper highlights these risks by introducing a new trajectory-matching model, ST-ATT, which leverages attention-enhanced Long Short-Term Memory (LSTM) to effectively capture the spatio-temporal correlations within trajectories. ST-ATT excels in identifying similar trajectories. To defend against linkage attacks on spatio-temporal trajectories, including advanced models like ST-ATT, we propose a novel Differential Privacy (DP) mechanism specifically designed to address the privacy risks. We reveal that the privacy budget and violation probability for each spatial point explicitly depend on earlier timestamps. The privacy budget can be flexibly redistributed between spatial and temporal domains without compromising overall privacy. This mechanism complies with DP, even when spatio-temporal points are reordered due to perturbation. Experiments show that ST-ATT can accurately identify spatio-temporal trajectories perturbed by the existing DP methods adding noise solely to the spatial domain. The proposed spatio-temporal DP mechanism resists ST-ATT, highlighting the need for considering spatio-temporal correlations to ensure robust privacy protection in spatio-temporal trajectories.

Index Terms—Spatio-temporal trajectory, differential privacy, LSTM, attention gate.

I. INTRODUCTION

A SPATIO-TEMPORAL trajectory refers to a sequence of spatio-temporal points that capture both spatial and temporal dimensions of movement. The widespread release of trajectory data exposes sensitive personal information. Linkage and reconstruction attacks can identify individuals from their trajectory history, increasing the risk of personal information

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identifiers	spatio temporal trajectory(Space disturbance)			
Shannon	-8.585676, 41.148522,64637	-8.585712, 41.148639,64652	-8.585685, 41.148855,64667	-8.58573, 41.148927,64682
Pseudo-identifiers	spatio temporal trajectory(clean trajectory)			
2HGUH6	-8.585456, 41.148515,64637	-8.585756, 41.148678,64652	-8.585705, 41.148889,64667	-8.585690, 41.148989,64682
6YUY89	-8.585739, 41.148558,61234	-8.58573, 41.148828,61249	-8.585721, 41.148972,61264	-8.586288, 41.149017,61279
3ERYU8	-8.613963, 41.141169,36902	-8.614125, 41.141124,36917	-8.615088, 41.140926,36932	-8.615277, 41.140818,36947

Fig. 1. Each cell contains three data points: longitude, latitude, and time (in seconds). One dataset is unperturbed; the other includes DP noise trajectories with pseudo-identifier information. Despite the perturbed spatial information, the temporal trajectories remain identical, allowing an attacker to identify the perturbed trajectories by matching the unchanged time information.

exposure from trajectory data. A substantial body of research [1], [2], [3] indicates that the efficiency and accuracy of computing trajectory similarity using deep learning have progressively improved. This allows for rapid matching of original trajectories within large datasets. The complexity of identifying trajectory similarities has been reduced from traditionally $\mathcal{O}(L^2)$ to $\mathcal{O}(L)$, significantly enhancing the success probability of linkage attacks and reducing their costs. L denotes the trajectory length.

A seminal empirical study on sparse mobility and transaction data (e.g., mobile phone call records and credit card metadata) [4] shows that, in such datasets, only four spatio-temporal points are sufficient to uniquely re-identify about 95% of users. To protect trajectory privacy, researchers have developed many privacy models for releasing trajectory data, such as k -anonymity [5], l diversity [6], [7], spatial and temporal generalization [8], fake data injection [9], and Differential Privacy (DP) [10], [11], [12], [13], [14].

Research challenges: Among the others, DP provides strict mathematical guarantees that prevent attackers from making precise inferences about individual data samples. For this reason, it has become increasingly popular. Current DP schemes for trajectories exhibit a few limitations:

Unprotected temporal sequences: Existing works on DP models of trajectory data, e.g., [11], [15], [16], [17], [18], have overlooked the protection of the timestamps associated with *spatial trajectories*, referred to as *temporal trajectories*. In many applications, temporal trajectory information is essential. For instance, in Uber's carpooling service, temporal trajectory

information directly determines whether two passengers can share a ride. When adding noise to perturb temporal trajectories, the order of timestamps can change, dramatically distorting the *spatio-temporal trajectories*. After the order change, a challenge arises in determining if the obfuscated temporal and spatial trajectories satisfy DP.

Spatio-temporal correlation overlooked: Another challenge of using DP to protect the privacy of spatio-temporal trajectories lies in the inherent spatio-temporal correlation of the trajectories. By analyzing the spatio-temporal correlation, adversaries can launch linkage attacks. In such attacks, adversaries can exploit the residual patterns or the presence of “signature” points in the obfuscated trajectory data to infer sensitive locations, such as a user’s home or workplace. The locations, distinctive in a user’s movement pattern in the spatial and temporal domains, can be pieced together to reconstruct original trajectories, undermining privacy.

Contribution of this study: To reveal the impact of unprotected temporal sequences on the privacy of spatio-temporal trajectories, we develop a novel and powerful spatio-temporal trajectory-matching model, ST-ATT. Built upon the Long Short-Term Memory (LSTM) framework, ST-ATT incorporates time embedding and attention gates to effectively capture the periodicity of temporal trajectories and the intricate spatio-temporal correlations within spatio-temporal trajectories.

To prevent such a powerful trajectory-matching model like ST-ATT from being misused to launch linkage attacks, we also propose a new DP mechanism to preserve the privacy of spatio-temporal trajectories. Specifically, we establish the privacy budget and violation probability of each spatial point. The inherent correlation is captured between the spatial and temporal aspects of a trajectory, revealing that the privacy budget is transferable between the two domains. Moreover, we prove that a spatio-temporal trajectory satisfying DP has any of its segments satisfying DP, even though some of the spatio-temporal points may swap their orders due to the perturbation in the temporal domain.

Experiments (Tables II and III) indicate that many existing DP methods [11], [14], [16], [19], [20], which add noise only to obfuscate spatial trajectories, do not adequately preserve the privacy of spatio-temporal trajectories. ST-ATT achieves exceptionally high recognition rates of the trajectories, much better than the existing trajectory-matching models. Even if noise is added to spatio-temporal trajectories, privacy protection is insufficient without considering the quantitative spatio-temporal relationships. Nonetheless, the experiments demonstrate that even the ST-ATT model cannot effectively detect spatio-temporal trajectories protected by our DP mechanism.

The key contributions of this paper are as follows:

- We design a new and powerful trajectory-matching model, ST-ATT, based on LSTM with time embedding and attention gates, which can effectively excavate the periodicity of temporal trajectories and the spatio-temporal correlations of trajectories.
- We propose a new DP mechanism for spatio-temporal trajectories, which balances the privacy budget and violation probability across spatial and temporal dimensions of trajectories. The DP mechanism ensures that any trajectory

TABLE I
NOTATION AND DEFINITION

Symbol	Description
L	The total number of points in trajectories, i.e., length.
$\mathcal{M}$	An ε -DP mechanism to protect privacy.
$f(\cdot)$	A query function retrieves or processes data from the database or dataset.
$p^{(i)}$	The i -th spatial (location) point in a trajectory, representing a specific geographic location.
$t^{(i)}$	The timestamp corresponding to the i -th spatial point $p^{(i)}$, indicating when the location data was recorded.
T^{st}	A sequence representing spatio-temporal trajectory points, defined as $((p^1, t^1), \dots, (p^L, t^L))$, capturing both spatial and temporal dimensions of movement.
$T^{st'}$	A perturbed version of T^{st} obtained by adding Gaussian noise, denoted as $((p^{1'}, t^{1'}), \dots, (p^{L'}, t^{L'}))$, ensuring that the data retains utility while protecting individual privacy.
T^{t_j}	A subset of the temporal component up to the j -th timestamp, defined as $(t^1, t^2, \dots, t^j)$. This represents a partial trajectory in the time dimension.

segment remains DP-compliant even if spatio-temporal points swap their order due to temporal perturbations.

- We prove analytically that the privacy budget can be transferred between spatial and temporal components without compromising overall privacy, allowing for flexible privacy preservation strategies.

As corroborated experimentally on two publicly available, real-world vehicular trajectory datasets, Porto and Geolife, ST-ATT can evade existing DP methods that preserve the privacy of spatio-temporal trajectories based solely on spatial obfuscation. However, ST-ATT fails to evade the proposed spatio-temporal DP mechanism, confirming the importance of considering spatial and temporal correlations in preserving the privacy of spatio-temporal trajectories. Our mechanism outperforms the existing protection methods in top- k [1], [2], [3], Mutual Information (MI) [21], and Trip Distribution (TE) [11] under the same privacy budget, thanks to the consideration of spatio-temporal correlations.

The rest of this paper is outlined as follows. Section II reviews the related work. Section III introduces theorems and definitions utilized in this paper and presents the system model. Section IV designs the ST-ATT model that can effectively detect similar trajectories by excavating spatio-temporal correlations. Section V proves the relationship between the privacy budget of spatial and temporal trajectory points, and proposes the new spatio-temporal DP mechanism. Section VI presents our experimental results. Section VII concludes this study and provides future outlooks. The key notation used in this paper is summarized in Table I.

II. RELATED WORK

A. Privacy Preservation for Trajectories

There are three approaches to trajectory protection using DP: One approach adds noise directly to the original trajectory points, another synthesizes the transition probabilities with noise, and the third is based on the frequency of perturbed trajectory points. Jiang et al. [22] developed three approaches to noise addition: targeting the complete trajectory, the sampling

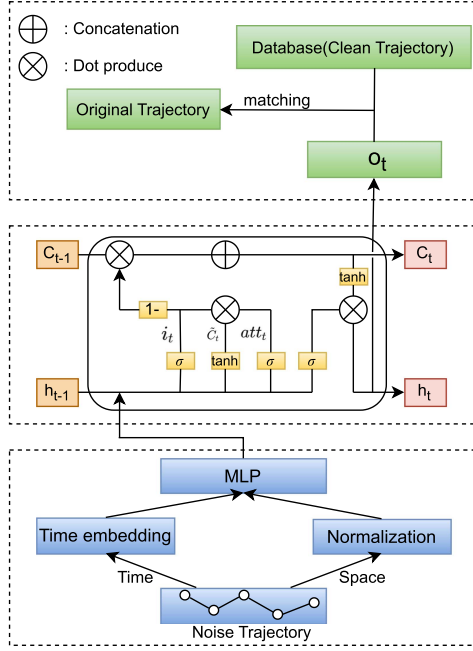


Fig. 2. The workflow of the model, where spatio-temporal trajectories are first processed through normalization and time embedding and input into an MLP network to be transformed into high-dimensional space vectors. Next, these high-dimensional vectors are fed into ST-ATT to extract the feature vectors of the trajectories. Finally, by calculating the distances between the feature vector of a perturbed trajectory and the feature vectors of unperturbed trajectories in the dataset, the most similar trajectory to the perturbed is identified.

points, and the trajectory point coordinates. They developed the Sampling Distance and Direction (SDD) mechanism to improve the transition between trajectory points, which uses exponential noise to sample distance and direction. Other DP models [23], [24], [25] incorporate various types of noise into original trajectories. These methods focus on the spatial aspects of trajectory noise and consider no temporal information.

DP Trajectory (DPT) [11] differs from the method of directly adding noise to the original trajectory. DPT models trajectories through a layered reference system and injects Laplacian noise into a prefix tree. The transition probabilities are distorted in compliance with DP, yet they still manage to represent the movement patterns of the original trajectories to a certain extent. The synthetic trajectories are generated from the noise tree, independent of real trajectories. Variants of DPT, e.g., DP-STAR [15], SPRT [16], LDPTTrace [19], and AdaTrace [7], consider additional information and address vulnerabilities of DPT, producing more private and useful trajectories. This method involves the synthesis of spatial trajectories and does not consider the protection of temporal trajectory information. Attackers can launch linkage attacks to match two trajectories with the same temporal trajectories from different datasets, exposing the user's privacy.

The third method involves synthesizing new trajectories with noise based on the frequency of perturbed trajectory points. For instance, Jin et al. [14] preserved personal privacy in the context of trajectory data collection from GPS-enabled devices, focusing on the trade-off between privacy protection and data utility. They highlighted the threat of recovery attacks, exposing

sensitive locations in a user's trajectory. They improved a DP-based model that employs randomization mechanisms to perturb the frequency distributions of key sensitive points and protect against re-identification attacks. Unfortunately, the model only alters the frequency of trajectories, resulting in the perturbed trajectory points still sharing strong similarities with the original points. Moreover, the model developed in [14] focuses solely on spatial trajectories and overlooks the temporal information. By overlooking temporal information, the method falls short of providing a robust solution for privacy protection in trajectory data.

Although the above studies protect privacy to a certain extent, they only consider spatial information and do not defend temporal trajectories. As shown in Fig. 1, the temporal trajectories remain unchanged while the spatial information is perturbed. This consistency enables attackers to re-identify perturbed trajectories by matching unchanged time information, rendering these methods ineffective to linkage attacks.

B. Trajectory Similarity Computation

Learning-based techniques have recently been developed to mitigate the high computational costs and lack of generalization inherent in non-learning methods [26], [27], [28]. Leveraging machine learning techniques, each trajectory is embedded into a vector in a multi-dimensional space. The trajectory similarity is converted to the similarity between two corresponding vectors, reducing the complexity. In addition, learning-based methods can generalize to different distance metrics. NeuTraj [3], T3S [11], Traj2SimVec [29], and TMN [2] are representative learning-based methods for similarity computation. These models focus on the similarity of spatial trajectories and fall short in handling the similarity of spatio-temporal trajectories, especially when considering interplay between the temporal and spatial aspects of trajectories. More intricate and sophisticated methods are required to calculate the similarity of spatio-temporal trajectories.

III. SYSTEM MODEL AND PROBLEM STATEMENT

In this section, we present the Sequential and the Parallel Composition theorems of DP and the definition of sensitivity, followed by the system model and problem statement.

Lemma 1 (Sequential Composition [30]): Assume a set of privacy mechanisms $\mathcal{M} = \{\mathcal{M}_1, \dots, \mathcal{M}_m\}$ gives ϵ_i -DP ($i = 1, \dots, m$). Then, $\mathcal{M}$ satisfies $(\sum_i \epsilon_i)$ -DP for this dataset.

Lemma 2 (Parallel Composition [30]): If there are two mechanisms yielding ϵ -DP, when they operate in parallel on two disjoint sub-datasets, the overall privacy loss is the more considerable privacy loss of the two. That is, their combination provides ϵ -DP. For k parallel ϵ -DP mechanisms operating on k disjoint sub-datasets, their combined effect remains ϵ -DP.

Definition 1 ((ϵ, δ) -DP): Let $\mathcal{M}$ be a randomized mechanism with domain $\mathcal{D}$. We say that $\mathcal{M}$ satisfies (ϵ, δ) -DP if, for every pair of neighboring datasets $D, D' \in \mathcal{D}$ and every measurable set S in the output space of $\mathcal{M}$, the following inequality holds:

$$\Pr[\mathcal{M}(D) \in S] \leq e^\epsilon \Pr[\mathcal{M}(D') \in S] + \delta.$$

In our paper, D and D' denote two neighboring datasets of spatio-temporal trajectories, where each data record corresponds to one complete spatio-temporal trajectory associated with an individual. D and D' are considered neighboring if they differ in at most one complete spatio-temporal trajectory, i.e., one trajectory is added, removed, or replaced. We set the slack parameter δ to 10^{-5} , which means that the above bound is allowed to be violated with only a very small probability.

Definition 2 (ℓ_2 -Sensitivity): Given a query function f , we define the ℓ_2 -sensitivity of f as when D and D' are different at most one complete spatio-temporal trajectory:

$$\Delta f = \max_{D, D'} \|f(D) - f(D')\|_2. \quad (1)$$

System Model: In this paper, given the training set of spatio-temporal trajectories, we calculate the sensitivity of both temporal and spatial trajectories according to this definition. The query function f represents the calculation of the Hausdorff distance. The sensitivity of our dataset is given by the maximum distance, normalized by the size of the dataset.

The considered system consists of two entities:

1) *Spatio-Temporal Trajectories:* The dataset collects spatio-temporal trajectory sequences of spatial and temporal points, as well as the trajectories of the movement of individuals in both the spatial and temporal domains.

2) *Query:* A querier can use deep learning models to perform similarity matching. As shown in Fig. 1, the querier may have a perturbed trajectory and wish to identify the corresponding trajectory from an unperturbed dataset to match the trajectory.

Problem Statement: To enhance the privacy protection of trajectories with both spatial and temporal features, we aim to address the challenge of resisting linkage attacks on spatio-temporal trajectories. In particular,

1) We want to validate the conjecture that unprotected temporal sequences of spatio-temporal trajectories can expose the trajectories to linkage attacks;

2) We wish to develop a mechanism to preserve the privacy of spatio-temporal trajectories against linkage attacks.

These two are addressed sequentially in Sections IV and V.

IV. ST-ATT: A NEW SPATIO-TEMPORAL TRAJECTORY MATCHING MODEL

While deep learning models are adept at handling sequence tasks, few deep learning models can effectively learn spatio-temporal correlations in the case of DP perturbations. We design a new detection model (a component designed to quantify the matching ratio between perturbed trajectories and their ground truth, assessing the privacy level ensured by the noise-injection mechanism), ST-ATT, as shown in Fig. 2; ST-ATT comprises a time embedding, a normalization, an MLP, and a variant LSTM network with an attention gate.

Given any spatio-temporal trajectory $T^{st} = [(p^1, t^1), \dots, (p^L, t^L)]$, the corresponding spatial and temporal trajectories are $T^s = [p^1, \dots, p^L]$ and $T^t = [t^1, \dots, t^L]$, respectively. ST-ATT finds the maximum length of all trajectories and makes all trajectories have consistent length by padding zeros to the

end. For spatial trajectory points, we normalize the coordinates, i.e., $\text{Nor}(p^i) = \frac{p^i - \mu}{\sigma}$, and then map the normalized into a higher-dimensional space using non-linear Multi-Layer Perceptron (MLP) layers $vs^i = \text{MLP}(\text{Nor}(p^i))$, where vs^i is a d -dimensional vector. The higher dimensional vector corresponding to T^s is $T_{vs} = [vs^1, \dots, vs^L]$.

The temporal trajectory is one-dimensional data. Time has periodic characteristics since people usually have the same daily behavior [31]. In our setting, this periodicity is used as a data-driven modeling prior based on the mobility datasets we study, rather than as a universal property of all spatio-temporal trajectories. To retain this periodic feature, we take one day as a cycle and convert time into two-dimensional data using the following function:

$$t_1^i = \frac{1}{2} \cos\left(\frac{t^i \cdot 2\pi}{86,400}\right) + \frac{1}{2}; \quad t_2^i = \frac{1}{2} \sin\left(\frac{t^i \cdot 2\pi}{86,400}\right) + \frac{1}{2}, \quad (2)$$

where 86,400 is the number of seconds in a day. T^t can be rewritten as $T^t = [(t_1^1, t_2^1), \dots, (t_1^L, t_2^L)]$. The periodicity of time is retained. Temporal information is enriched.

To obtain high-dimensional temporal trajectories, like spatial trajectories, we use a non-linear layer to process the two-dimensional temporal trajectories, i.e., $vt^i = \text{MLP}(t_1^i, t_2^i)$, which is d -dimensional. The higher-dimensional vector of T^t is $T_{vt} = [vt^1, \dots, vt^L]$.

A. Attention Mechanism

The non-linear MLP network allows us to obtain the spatial trajectory embedding, denoted as $T^{vs} = [vs^1, \dots, vs^L]$, and the temporal trajectory embedding, denoted as $T^{vt} = [vt^1, \dots, vt^L]$. We design a pairwise attention preprocessing model to capture the internal connections in the spatio-temporal trajectory and the spatio-temporal correlations between spatial and temporal trajectories. This model captures complete trajectory association information and can more accurately identify the original trajectories corresponding to DP-perturbed trajectories, as follows:

- 1) The self-attention of a spatial trajectory to another spatial trajectory is given by $\text{Att}_{s \rightarrow s} = \frac{\exp(W_1 T^{vs} \odot W_2 (T^{vs})^T)}{\sum \exp(W_1 T^{vs} \odot W_2 (T^{vs})^T)}$;
- 2) The self-attention of a temporal trajectory to another temporal trajectory is given by $\text{Att}_{t \rightarrow t} = \frac{\exp(W_1 T^{vt} \odot W_2 (T^{vt})^T)}{\sum \exp(W_1 T^{vt} \odot W_2 (T^{vt})^T)}$;
- 3) The attention of a spatial trajectory to another temporal trajectory is given by $\text{Att}_{s \rightarrow t} = \frac{\exp(W_1 T^{vs} \odot W_2 (T^{vt})^T)}{\sum \exp(W_1 T^{vs} \odot W_2 (T^{vt})^T)}$;
- 4) The attention of a temporal trajectory to another spatial trajectory is given by $\text{Att}_{t \rightarrow s} = \frac{\exp(W_1 T^{vt} \odot W_2 (T^{vs})^T)}{\sum \exp(W_1 T^{vt} \odot W_2 (T^{vs})^T)}$.

where W_1 and W_2 are parameter matrices used to enhance the representation of the trajectory. $(T^{vs})^T$ and $(T^{vt})^T$ represent the transposes of T^{vs} and T^{vt} , respectively.

We obtain the higher-dimensional space vector representation of the new spatial and temporal trajectories, T'_{vs} and T'_{vt} :

$$T'_{vs} = 2 \cdot T_{vs} - \text{Att}_{s \rightarrow s} \cdot T_{vs}^T - \text{Att}_{t \rightarrow s} \cdot T_{vt}^T; \quad (3)$$

$$T'_{vt} = 2 \cdot T_{vt} - \text{Att}_{t \rightarrow t} \cdot T_{vt}^T - \text{Att}_{s \rightarrow t} \cdot T_{vs}^T. \quad (4)$$

Here, T'_{vt} and T'_{vs} contain richer information on internal trajectory and spatio-temporal associations; cf. T_{vt} and T_{vs} .

TABLE II
PERFORMANCE OF FULL ATTENTION (IN %)

	Geolife					Porto				
	HR@1	HR@5	HR@10	HR@20	HR@50	HR@1	HR@5	HR@10	HR@20	HR@50
S	5.67	11.43	16.84	24.78	34.91	8.75	19.1	27.85	32.85	42.04
ST	1.9	5.92	9.84	13.89	19.62	3.5	9.05	16.87	22.87	32.75
F [14]	6.16	13.79	18.41	23.53	30.97	10.89	25.47	29.88	34.69	45.76
DPT [11]	4.12	9.56	13.89	18.72	25.98	7.56	17.45	25.89	30.88	39.78
SPRT [16]	2.87	5.67	9.34	13.79	19.62	5.15	10.56	18.81	21.78	28.98
LDPTTrace [19]	2.97	6.32	10.12	14.65	20.62	4.18	9.66	15.77	19.23	25.39
RetraSyn [20]	2.09	5.21	8.68	12.81	17.50	3.59	7.94	13.71	16.39	21.37
ST-Cor	0.45	1.01	1.86	2.88	4.78	0.7	1.6	2.9	4.32	6.74

B. LSTM With Attention Gate

We propose a new attention-based LSTM model instead of simply using the original LSTM model. Specifically, we design a new attention gate to learn the relationships within and between spatial and temporal trajectories. For each time step, the attention gate can retain more connections within the trajectories, increase the spatio-temporal relationships, and filter out less relevant information. At the same time, the attention vector can highlight important information, which is essential information saved in the current hidden layer.

1) *Coupled LSTM*: To better describe the structure of ST-ATT, we first review the structure of coupled LSTM [32], which is a variant of LSTM, coupling the input and forget gates of LSTM. It improves the correlation between the input gate and the forget gate, with the following structure:

$$i_t = \sigma_i(W_i[h_{t-1}, x_t] + b_i); \quad (5)$$

$$\tilde{c}_t = \tanh(W_c[h_{t-1}, x_t] + b_c); \quad (6)$$

$$o_t = \sigma_o(W_o[h_{t-1}, x_t] + b_o); \quad (7)$$

$$c_t = (1 - i_t) * c_{t-1} + i_t * \tilde{c}_t; \quad (8)$$

$$h_t = o_t * \tanh(c_t), \quad (9)$$

where W_i, W_f, W_c , and W_o are weight parameters that connect different inputs and gates with storage units and outputs; b_i, b_f, b_c , and b_o are the corresponding biases.

2) *Proposed Attention LSTM*: Although outperforming LSTM, coupled LSTM overlooks the correlation between temporal and spatial trajectories. To capture temporal and spatial information and their correlation, we design ST-ATT to replace the original coupled LSTM. We create an attention gate to control the influence of attention outcomes on trajectory representations. For each trajectory point, the attention gate can select critical context information that affects the current trajectory point according to the weight results of attention.

According to the update process of coupled LSTM, an attention gate is added to each unit of coupled LSTM. The spatial and temporal trajectory embedding and the attention calculate the result of the attention gate as

$$att_t = \sigma_{att}(W_{att}[x_t \oplus x'_t] + b_{att}), \quad (10)$$

where $(\cdot \oplus \cdot)$ denotes concatenation; W_{att} and b_{att} are the weight and bias of the attention gate, respectively; x_t is spatial and temporal trajectory embedding, i.e., $x_t = \lambda \cdot T_{vs} + (1 - \lambda) \cdot T_{vt}$ and x'_t is the attention result, i.e., $x'_t = \lambda \cdot T'_{vs} + (1 - \lambda) \cdot T'_{vt}$.

Accordingly, we need to reflect on the influence of the attention gate on the long memory and output gate. Hence, we replace (7) and (8) with

$$c_t = (1 - i_t) * c_{t-1} + i_t * att_t * \tilde{c}_t; \quad (11)$$

$$o_t = \sigma_o(W_o[h_{t-1}, x_t \oplus x'_t] + b_o), \quad (12)$$

where o_t is the high-dimensional spatial vector representation of the spatio-temporal trajectories.

C. Training Process and Prediction Process

1) *Training Process*: We use the original definition of the Hausdorff distance to calculate the distance matrix between each perturbed trajectory and each unperturbed trajectory, treating it as the ground truth. Using the ST-ATT model, all trajectories can be transformed from their representations of latitude, longitude, and time into high-dimensional vector representations. By calculating the Euclidean distances between these high-dimensional vectors, a predicted value is obtained. In our formulation, the ground-truth value $D_{\text{true}}^{(i)}$ corresponds to the true Hausdorff distance between the *original* (unperturbed) trajectories, while the predicted value $D_{\text{predicted}}^{(i)}$ corresponds to the distance computed between the *perturbed* trajectories in the learned embedding space.

Since approximating trajectory similarity is a regression problem, we adopt the Root Mean Squared Error (RMSE) as the loss function, which aims to minimize the distance between the predicted and the ground truth for each perturbed and unperturbed trajectory, as given by $L_{\text{global}} = \sqrt{\frac{1}{N} \sum_{i=1}^N (D_{\text{true}}^{(i)} - D_{\text{predicted}}^{(i)})^2}$. Inspired by [29], we apply data augmentation to the training set by incorporating a large amount of sub-trajectory information. This enables the model to achieve better training performance with fewer trajectory training data. Similarly, the goal here is also to minimize the gap between the predicted and true values for the sub-trajectories: $L_{\text{sub}} = \sqrt{\frac{1}{M} \sum_{j=1}^M (d_{\text{true}}^{(j)} - d_{\text{predicted}}^{(j)})^2}$, where N and M are the numbers of trajectory pairs and sub-trajectory pairs, respectively; $D_{\text{true}}^{(i)}$, $D_{\text{predicted}}^{(i)}$, $d_{\text{true}}^{(j)}$, and $d_{\text{predicted}}^{(j)}$ represent the ground-truth and predicted distances for the full trajectories and sub-trajectories, respectively.

To jointly optimize the global and local (sub-trajectory) similarity, we introduce a weighting coefficient α and combine the two loss terms as $L = \alpha L_{\text{global}} + (1 - \alpha) L_{\text{sub}}$. By default, we set $\alpha = 0.8$ so that the global trajectory structure receives a higher priority. This choice is motivated by the fact that global trajectory similarity contributes more to accurate trajectory

matching, while sub-trajectory information serves primarily as auxiliary fine-grained supervision.

2) *Prediction Process*: Using the trained ST-ATT model, we can transform trajectories from representations of latitude, longitude, and time into high-dimensional representations. The pairwise distance between all perturbed trajectories and unperturbed trajectories is calculated and sorted from the most similar to the least similar. If only the spatial trajectories are perturbed, we can identify the original trajectory of a perturbed trajectory by finding the trajectory in the unperturbed dataset whose temporal trajectory has a distance of zero from the perturbed temporal trajectory. If both the spatial and temporal trajectories are perturbed, the trajectory in the unperturbed dataset with the smallest distance to the perturbed trajectory is considered the original trajectory.

V. NEW SPATIO-TEMPORAL DP MECHANISM FOR TRAJECTORY PRIVACY

A. Spatio-Temporal DP Mechanism

We analyze the privacy of the spatio-temporal DP mechanisms, where each spatio-temporal point of a trajectory is independently perturbed using the Gaussian mechanism. We first introduce two important lemmas. The first is extended from the Azuma inequality [33]. The second utilizes Taylor expansion to bound the expected privacy loss.

Lemma 3: Consider k random variables, i.e., $C_1, \dots, C_i, \dots, C_k$. For any random variable C_i , if $\mathbb{E}[C_i|C_1 = c_1, \dots, C_{i-1} = c_{i-1}] \leq \beta_i$ and $|C_i| < \alpha_i$, then $\Pr[\sum_{i=1}^k C_i > \sum_{i=1}^k \beta_i + z\sqrt{\sum_{i=1}^k \alpha_i^2}] \leq e^{-z^2/2}$, where $z > 0$ is any positive real number.

Lemma 4: Let $\mathcal{M}$ be a randomized mechanism that satisfies ε -DP. Suppose that D and D' are two neighboring datasets with output distributions P_X and P_Y , respectively, where $X := \mathcal{M}(D)$ and $Y := \mathcal{M}(D')$. The expectation of the log-likelihood ratio is bounded by $\mathbb{E}[\ln \frac{\Pr[X]}{\Pr[Y]}] = D_{KL}(P_X \| P_Y) \leq e^\varepsilon - 1$, where $D_{KL}(\cdot \| \cdot)$ is the Kullback-Leibler (KL) divergence and the probabilities concern the output distributions of $\mathcal{M}$ on D and D' .

Proof: By the definition of the KL divergence, we have

$$D_{KL}(P_X \| P_Y) = \sum_t P_X(t) \ln \frac{P_X(t)}{P_Y(t)} = \mathbb{E} \left[\ln \frac{\Pr[X]}{\Pr[Y]} \right],$$

Since $\mathcal{M}$ is ε -DP, for all measurable events S and any pair of adjacent datasets (D, D') , we have $e^{-\varepsilon} \leq \frac{\Pr[\mathcal{M}(D) \in S]}{\Pr[\mathcal{M}(D') \in S]} \leq e^\varepsilon$. Taking $S = \{t\}$ (or using the corresponding density in the continuous case), we obtain for every possible output t that $e^{-\varepsilon} \leq \frac{P_X(t)}{P_Y(t)} \leq e^\varepsilon$. Define the likelihood ratio $r(t) := \frac{P_X(t)}{P_Y(t)}$. Then $r(t) \in [e^{-\varepsilon}, e^\varepsilon]$, $\forall t$. Using the elementary inequality $\ln x \leq x - 1$, $\forall x > 0$, we obtain $D_{KL}(P_X \| P_Y) = \sum_t P_X(t) \ln r(t) \leq \sum_t P_X(t) (r(t) - 1) = \sum_t (P_X(t)^2 / P_Y(t) - P_X(t))$. From the DP constraint, we also have $P_X(t) \leq e^\varepsilon P_Y(t)$, or $\frac{P_X(t)^2}{P_Y(t)} \leq e^\varepsilon P_X(t)$, $\forall t$.

Therefore, $D_{KL}(P_X \| P_Y) \leq \sum_t (e^\varepsilon P_X(t) - P_X(t)) = e^\varepsilon \sum_t P_X(t) - 1 = e^\varepsilon - 1$, which completes the proof. $\square$

Given Lemmas 3 and 4, we present the main finding of this paper, that is, the privacy budget of spatial trajectory points is related only to the privacy budget of their associated temporal points and the probability of the spatial point violating DP, as established in the following theorem.

Theorem 1: Consider the spatio-temporal correlation of the whole trajectory. Given a spatio-temporal trajectory, denoted by $T^{st} = [(p^1, t^1), (p^2, t^2), \dots, (p^L, t^L)]$, if the spatial trajectories p^j satisfies $(\varepsilon_{p_j}, \delta_{p_j})$ -DP, and the temporal trajectories t^j satisfies ε_{t_j} -DP, then the spatial trajectory point p^j satisfies

$$\begin{cases} \varepsilon_{p_j} = \sum_{i=1}^j (e^{\varepsilon_{t_i}} - 1) + \sqrt{2 \ln(1/\delta_{p_j}) \sum_{i=1}^j \varepsilon_{t_i}^2}; \\ \delta_{p_j} = e^{-z^2/2}, z > 0. \end{cases} \quad (13)$$

where $z > 0$ is any positive real number.

Proof: Given any spatio-temporal trajectory $T^{st} = [(p^1, t^1), \dots, (p^L, t^L)]$, the spatial and temporal components of this trajectory are $T^s = [p^1, \dots, p^L]$ and $T^t = [t^1, \dots, t^L]$, respectively. Consider a relational function $\phi(\cdot)$ that captures the spatial and temporal dependencies. The function describes how the position of a point changes over time. For any point $p^j \in T^s$, $j \in [1, L]$, the spatial and temporal relationship can be defined as

$$p^j = \phi(T^{t_j}), \quad (14)$$

where $T^{t_j} = [t^1, \dots, t^j] \subseteq T^t$ is related to p^j . In addition, various factors affect the spatial point p^j , such as other spatial trajectory points related to p^j . Since we only consider the spatial and temporal relationship in $\phi(\cdot)$, we consider the effect of T^{t_j} and treat other impacts as constants.

Given a perturbed spatio-temporal trajectory $T^{st'} = [(p^{1'}, t^{1'}), (p^{2'}, t^{2'}), \dots, (p^{L'}, t^{L'})]$, and a DP-based query function $f(\cdot)$, based on the definition of DP, each spatial and temporal point on the trajectory satisfies

$$\begin{cases} \Pr[f(p^j) = P^j | A_j] \leq e^{\varepsilon_{p_j}} \Pr[f(p^{j'}) = P^j | A_j'] + \delta_{p_j}; \\ \Pr[f(t^j) = T^j | B_j] \leq e^{\varepsilon_{t_j}} \Pr[f(t^{j'}) = T^j | B_j']. \end{cases} \quad (15)$$

where P^j and T^j are the possible values of p^j and t^j obtained by the function $f(\cdot)$, respectively. A_j , B_j , A_j' , and B_j' are the factors affecting p^j , t^j , $p^{j'}$ and $t^{j'}$, respectively. The probability of $f(p^j) = P^j$ is $\Pr[f(p^j) = P^j | A_j]$. The probability of $f(t^j) = T^j$ is $\Pr[f(t^j) = T^j | B_j]$. Equation (15) indicates every spatial trajectory point p^j and its temporal trajectory point t^j satisfy the $(\varepsilon_{p_j}, \delta_{p_j})$ -DP and ε_{t_j} -DP, respectively.

From the dependency between the spatial and temporal trajectory points in (14), we have $p^j = \phi(T^{t_j})$ and $p^{j'} = \phi(T^{t_j'})$. Suppose that there exists Φ satisfy $\Phi(p^j) = \phi(f(T^{t_j}))$ and $\Phi(p^{j'}) = \phi(f(T^{t_j'}))$. Then, we obtain

$$\begin{cases} \Pr[\Phi(p^j) = P^j] = \Pr[\phi(f(T^{t_j})) = P^j]; \\ \Pr[\Phi(p^{j'}) = P^j] = \Pr[\phi(f(T^{t_j'})) = P^j]. \end{cases} \quad (16)$$

According to the chain rule, (16) is rewritten as

$$\begin{aligned} \Pr[\Phi(p^j) = P^j] &= \Pr[\phi(f(T^{t_j})) = P^j] \\ &= \Pr[f(t^1) = T^1, \dots, f(t^j) = T^j] = \prod_{i=1}^j \Pr[f(t^i) = T^i | B_i], \end{aligned} \quad (17)$$

where $B_i = [f(t^1) = T^1, \dots, f(t^{i-1}) = T^{i-1}]$. Similarly,

$$\Pr[\Phi(p^{j'}) = P^j] = \prod_{i=1}^j \Pr[f(t^i) = T^i | B'_i], \quad (18)$$

where $B'_i = [f(t^1) = T^1, \dots, f(t^{i-1}) = T^{i-1}]$.

For the brevity of notation, we define K_i that satisfies the following relationship as

$$K_i = \ln \left(\frac{\Pr[f(t^i) = T^i | B_i]}{\Pr[f(t^i) = T^i | B'_i]} \right), \quad (19)$$

which is the logarithm of the probability ratio that both $f(t^i)$ and $f(t^{i'})$ output T^i . Plugging (17) and (18) in (19) yields

$$\begin{aligned} \sum_{i=1}^j K_i &= \ln \left(\frac{\Pr[f(t^1) = T^1, \dots, f(t^j) = T^j]}{\Pr[f(t^1) = T^1, \dots, f(t^j) = T^j]} \right) \\ &= \ln \left(\frac{\Pr[\Phi(p^j) = P^j]}{\Pr[\Phi(p^{j'}) = P^j]} \right). \end{aligned} \quad (20)$$

Since $f(\cdot)$ satisfies DP, it readily follows that

$$|K_i| = \left| \ln \left(\frac{\Pr[f(t^i) = T^i | B_i]}{\Pr[f(t^i) = T^i | B'_i]} \right) \right| \leq \varepsilon_{t_i}. \quad (21)$$

For conciseness, we define $M = (f(t^i) = T^i | B_i)$ and $N = (f(t^{i'}) = T^i | B'_i)$. By Lemma 4, the expectation of K_i is

$$\begin{aligned} \mathbb{E}[K_i | K_1 = k_1, \dots, K_{i-1} = k_{i-1}] \\ = \mathbb{E} \left[\ln \left(\frac{\Pr[f(t^i) = T^i | B_i]}{\Pr[f(t^i) = T^i | B'_i]} \right) \right] \leq e^{\varepsilon_{t_i}} - 1. \end{aligned} \quad (22)$$

According to (21), (22) and Lemma 3, we have

$$\Pr \left[\sum_{i=1}^j K_i > \sum_{i=1}^j (e^{\varepsilon_{t_i}} - 1) + z \sqrt{\sum_{i=1}^j \varepsilon_{t_i}^2} \right] \leq e^{-z^2/2}, \quad (23)$$

where $z > 0$ is any positive real number.

Define $\varepsilon_{p_j} = \sum_{i=1}^j (e^{\varepsilon_{t_i}} - 1) + \sqrt{2 \ln(1/\delta_{p_j}) \sum_{i=1}^j \varepsilon_{t_i}^2}$ and $\delta_{p_j} = e^{-z^2/2}$. Then, (23) can be rewritten as $\Pr[\sum_{i=1}^j K_i > \varepsilon_{p_j}] \leq \delta_{p_j}$. According to (20), we have

$$\Pr \left[\frac{\Pr[\Phi(p^j) = P^j]}{\Pr[\Phi(p^{j'}) = P^j]} > e^{\varepsilon_{p_j}} \right] \leq \delta_{p_j}. \quad (24)$$

In other words, if we consider the relationship between the spatial and temporal domains, the spatial trajectory point p^j satisfies $(\sum_{i=1}^j (e^{\varepsilon_{t_i}} - 1) + \sqrt{2 \ln(1/\delta_{p_j}) \sum_{i=1}^j \varepsilon_{t_i}^2}, e^{-z^2/2})$ -DP. This proof is complete. $\square$

From (24), the privacy budget of spatial point p^j is only related to the privacy budgets of the associated temporal points and the probability of this point violating the privacy budget. On the other hand, the privacy budget of a temporal point t^j cannot be represented by the privacy budgets of its associated spatial points since, in a trajectory, each temporal point corresponds to a unique spatial point. However, a spatial point may correspond to multiple temporal points.

Theorem 1 considers the spatio-temporal correlations across a trajectory, and can incur high computational overhead for long trajectories. We proceed to consider local spatio-temporal correlations with improved efficiency.

Theorem 2: Given a spatio-temporal trajectory, $T^{st} = ((p^1, t^1), \dots, (p^L, t^L))$, if the spatial trajectory point p^j satisfies $(\varepsilon_{p_j}, \delta_{p_j})$ -DP and the temporal trajectory point t^j satisfies ε_{t_j} -DP, considering the impact of the D points before the j -th point on the point, the spatial trajectory point p^j satisfies

$$\begin{cases} \varepsilon_{p_j} = \sum_{i=j-D}^j (e^{\varepsilon_{t_i}} - 1) + \sqrt{2 \ln(1/\delta_{p_j}) \sum_{i=j-D}^j \varepsilon_{t_i}^2}; \\ \delta_{p_j} = e^{-z^2/2}, z > 0. \end{cases} \quad (25)$$

where $z > 0$ is any positive real number.

Proof: We focus locally on the sliding window around the trajectory point p^j , which is differentiable. In other words, we consider the range $T_W^{stj} = [(p^{j-D}, t^{j-D}), \dots, (p^j, t^j)]$ with D being the sliding window size. The spatial sliding window trajectory and temporal sliding window trajectory are $T_W^{sj} = [p^{j-D}, \dots, p^j]$ and $T_W^{tj} = [t^{j-D}, \dots, t^j]$, respectively. The advantage of this method is that it avoids the expensive computation caused by considering the whole trajectory and can focus on important local information.

Similar to (14), we obtain $p_j = \phi(T_W^{tj})$, where T_W^{tj} is related to p^j and it is the subset of T^{tj} , i.e. $T_W^{tj} \subseteq T^{tj}$. Suppose there exists Φ satisfy $\Phi(p^j) = \phi(f(T_W^{tj}))$ and $\Phi(p^{j'}) = \phi(f(T_W^{t'j}))$. Like (16), we obtain

$$\begin{cases} \Pr[\Phi(p^j) = P^j] = \Pr[\phi(f(T_W^{tj})) = P^j]; \\ \Pr[\Phi(p^{j'}) = P^j] = \Pr[\phi(f(T_W^{t'j})) = P^j]. \end{cases} \quad (26)$$

Similar to (17) and (18), we have

$$\Pr[\Phi(p^j) = P^j] = \prod_{i=t-D}^j \Pr[f(t^i) = T^i | B_i]; \quad (27)$$

$$\Pr[\Phi(p^{j'}) = P^j] = \prod_{i=t-D}^j \Pr[f(t^i) = T^i | B'_i], \quad (28)$$

where $B_i = [f(t^{j-D}) = T^{j-D}, \dots, f(t^{i-1}) = T^{i-1}]$ and $B'_i = [f(t^{(j-D)'}) = T^{j-D}, \dots, f(t^{(i-1)'}) = T^{i-1}]$.

As opposed to considering the whole trajectory, each point $t^i \in T^{tj}$ can have corresponding perturbation point $t^{i'} \in T^{t'j}$. With the sliding window, a point $t^{i'} \in T^{t'j} - T_W^{t'j}$ can be outside the sliding window, and the point t^i corresponding to the point $t^{k'} \in T_W^{t'j}$ is obtained by the disturbance of point $t^k \in T^{tj} - T_W^{tj}$ outside the sliding window. We consider the following two cases:

Case 1: If the disturbance point $t^{i'}$ corresponding to the original trajectory t^i falls within the sliding window, that is, $t^{i'} \in T_W^{t'j}$, the i -th temporal trajectory point satisfies ε_{t_i} -DP:

$$\Pr[f(t^i) = T^i | B_i] \leq e^{\varepsilon_{t_i}} \Pr[f(t^{i'}) = T^i | B'_i]. \quad (29)$$

Case 2: If the disturbance point $t^{i'}$ corresponding to the original trajectory t_i does not fall within the sliding window, that is, $t^{i'} \in (T^{t'j} - T_W^{t'j})$, while the perturbation point $t^{(k')}$ corresponding to the position t_i is the perturbation of $t^{(k)} \in (T^{tj} - T_W^{tj})$, we can prove that the i -th temporal trajectory point satisfies ε_{t_i} -DP. According to Definition 2, we have

$$\|f(t^i) - f(t^{i'})\|_1 \leq \Delta f, \quad (30)$$

where Δf is the sensitivity, and $f(\cdot)$ is the query function.

Assume that the function $f(\cdot)$ is L_c -Lipschitz continuous, i.e., $\|f(t^i) - f(t^k)\| \leq L_c \|t^i - t^k\|$, $\forall t^i, t^k$, $L_c > 0$ and $t^i - t^k =$

Δt . By the definition of DP and (30), we have

$$\frac{\Pr[f(t^i) + N(0, \sigma^2) = T^i]}{\Pr[f(t^k) + N(0, \sigma^2) = T^i]} \leq \frac{\Pr[f(t^i) + N(0, \sigma^2) = T^i]}{\Pr[f(t^k) - \Delta f + N(0, \sigma^2) = T^i]} \quad (31a)$$

$$\leq \frac{\Pr[f(t^i) + N(0, \sigma^2) = T^i]}{\Pr[f(t^i) + L_c \Delta t - \Delta f + N(0, \sigma^2) = T^i]} \quad (31b)$$

$$= \frac{\Pr[N(0, \sigma^2) = T^i - f(t^i)]}{\Pr[f(N(0, \sigma^2) = T^i - f(t^i) + \Delta f - L_c \Delta t)} \quad (31c)$$

$$= \frac{e^{-\frac{(T^i - f(t^i))^2}{2\sigma^2}}}{e^{-\frac{(T^i - f(t^i) + \Delta f - L_c \Delta t)^2}{2\sigma^2}}} \leq \frac{e^{-\frac{(T^i - f(t^i))^2}{2\sigma^2}}}{e^{-\frac{(T^i - f(t^i) + \Delta f)^2}{2\sigma^2}}} \quad (31d)$$

$$= \frac{\Pr[f(t^i) + N(0, \sigma^2) = T^i]}{\Pr[f(t^i) - \Delta f + N(0, \sigma^2) = T^i]} \leq \varepsilon_{t_i}. \quad (31e)$$

where (31a) is obtained by substituting (30); (31b) is due to the L_c -Lipschitz continuity of the function $f(\cdot)$ and $t^i - t^k = \Delta t$; (31c) is attained by substituting (30) into (31b). We use the definition of Gaussian distribution to derive (31d) from (31c) and the definition of DP to get (31e). $\square$

Define $K_i = \ln\left(\frac{\Pr[f(t^i) = T^i | B_i]}{\Pr[f(t^i) = T^i | B'_i]}\right)$. As with (19) and (20),

$$\sum_{i=j-D}^j K_i = \ln\left(\frac{\Pr[\Phi(p^j) = P^j]}{\Pr[\Phi(p^j) = P^i]}\right), \quad (32)$$

where $t^{j'}$ is the disturbance point corresponding to point t^i in the sliding window, and it is not necessarily a direct perturbation of t^i . Similar to (21), (22) and (23), we have

$$|K_i| = \left| \ln\left(\frac{\Pr[f(t^i) = T^i | B_i]}{\Pr[f(t^{j'}) = T^j | B'_i]}\right) \right| \leq \varepsilon_{t_i}; \quad (33)$$

$$\begin{aligned} & \mathbb{E}[K_i | K_{t-D} = k_{t-D}, \dots, K_{i-1} = k_{i-1}] \\ &= \mathbb{E}\left[\ln\left(\frac{\Pr[f(t^i) = T^i | B_i]}{\Pr[f(t^{j'}) = T^j | B'_i]}\right)\right] \leq e^{\varepsilon_{t_i}} - 1; \end{aligned} \quad (34)$$

$$\Pr\left[\sum_{i=t-D}^j K_i > \sum_{i=t-D}^j (e^{\varepsilon_{t_i}} - 1) + z\sqrt{\sum_{i=t-D}^j \varepsilon_{t_i}^2}\right] \leq e^{-\frac{z^2}{2}}. \quad (35)$$

Define $\varepsilon_{p_j} = \sum_{i=t-D}^j (e^{\varepsilon_{t_i}} - 1) + z\sqrt{\sum_{i=t-D}^j \varepsilon_{t_i}^2}$ and $\delta_{p_j} = e^{-z^2/2}$. Then, (35) can be written as $\Pr[\sum_{i=t-D}^j K_i > \varepsilon_{p_j}] \leq \delta_{p_j}$. According to (32), we have $\Pr[\frac{\Pr[\Phi(p^j) = P^j]}{\Pr[\Phi(p^j) = P^i]} > e^{\varepsilon_{p_j}}] \leq \delta_{p_j}$; i.e., the spatial trajectory point p^j satisfies $(\sum_{i=t-D}^j (e^{\varepsilon_{t_i}} - 1) + z\sqrt{\sum_{i=t-D}^j \varepsilon_{t_i}^2}, e^{-z^2/2})$ -DP. The spatio-temporal correlation of an entire trajectory incurs significant spatial and temporal complexity. We opt for local spatio-temporal correlations. $\square$

Algorithm 1 summarizes the allocation of the privacy budget for each spatio-temporal point of a trajectory given the overall privacy budget of the trajectory, i.e., ϵ . For illustration convenience, we assume that all temporal trajectory points are associated with equal privacy budgets: $\varepsilon_{t_1} = \dots = \varepsilon_{t_L}$. More adaptive assignment of privacy budgets is beyond the scope of this paper, and will be studied in our subsequent work.

Algorithm 1: Calculation of ε_{p_j} and ε_{t_i} .

Input: ϵ (The privacy budget for the trajectory); L (trajectory length); $\delta_{p_j} = 10^{-5}$, $\forall j$; set

$\varepsilon_{t_i} = \varepsilon_t$, $\forall i = 1, \dots, L$.

Output: ε_{p_j} (list of solutions) and ε_{t_i} .

1: **for** $j \leftarrow 1$ **to** L **do**

2: Compute ε_{p_j} in terms of ε_t using (25).

3: **end for**

4: Solve the equation $\sum_{j=1}^L \varepsilon_{p_j} + L \cdot \varepsilon_t = \epsilon$, and obtain ε_t .

5: Initialize an empty list ε_{p_j} to store computed privacy budgets for each trajectory point.

6: **for** $j \leftarrow 1$ **to** L **do**

7: Append ε_{p_j} to ε_{p_j} .

8: **end for**

9: **return** ε_{p_j} and ε_t .

B. Privacy Analysis

Each spatio-temporal point is perturbed by the Gaussian mechanism, which guarantees point-wise (ϵ_j, δ_j) -DP. By applying sequential composition to all perturbed points, the whole released trajectory satisfies trajectory-level DP, as established in the following theorem.

Theorem 3 (Trajectory Privacy Guarantee): A randomized mechanism $\mathcal{M}$ that independently perturbs each spatio-temporal point j with (ϵ_j, δ_j) -DP Gaussian noise preserves trajectory-level (ϵ, δ) -DP, provided that $\sum_{j=1}^L \epsilon_j \leq \epsilon$ and $\sum_{j=1}^L \delta_j \leq \delta$.

Proof: Assume that D and D' are two adjacent datasets differing in at most one complete spatio-temporal trajectory. Let $\mathcal{M}(D) = (\mathcal{M}_1(D), \dots, \mathcal{M}_L(D))$ denote the released perturbed trajectory, where $\mathcal{M}_j$ is the Gaussian mechanism applied to the j -th spatio-temporal point. For any measurable output $s = (s_1, \dots, s_L)$, $\Pr[\mathcal{M}(D) = s] = \prod_{j=1}^L \Pr[\mathcal{M}_j(D) = s_j]$ and $\Pr[\mathcal{M}(D') = s] = \prod_{j=1}^L \Pr[\mathcal{M}_j(D') = s_j]$, since the L point-wise mechanisms are independent. As a result, $\frac{\Pr[\mathcal{M}(D)=s]}{\Pr[\mathcal{M}(D')=s]} = \prod_{j=1}^L \frac{\Pr[\mathcal{M}_j(D)=s_j]}{\Pr[\mathcal{M}_j(D')=s_j]}$. Since each $\mathcal{M}_j$ satisfies (ϵ_j, δ_j) -DP, it follows that $\frac{\Pr[\mathcal{M}_j(D)=s_j]}{\Pr[\mathcal{M}_j(D')=s_j]} \leq e^{\epsilon_j}$ and, in turn, $\frac{\Pr[\mathcal{M}(D)=s]}{\Pr[\mathcal{M}(D')=s]} \leq \prod_{j=1}^L e^{\epsilon_j} = e^{\sum_{j=1}^L \epsilon_j}$. Meanwhile, by the sequential composition property, the failure probabilities accumulate as $\sum_{j=1}^L \delta_j$. Hence, the composed mechanism $\mathcal{M}$ satisfies $\Pr[\mathcal{M}(D) \in \mathcal{S}] \leq e^{\sum_{j=1}^L \epsilon_j} \Pr[\mathcal{M}(D') \in \mathcal{S}] + \sum_{j=1}^L \delta_j$ for any measurable output set $\mathcal{S}$. If $\sum_{j=1}^L \epsilon_j \leq \epsilon$ and $\sum_{j=1}^L \delta_j \leq \delta$, then $\Pr[\mathcal{M}(D) \in \mathcal{S}] \leq e^\epsilon \Pr[\mathcal{M}(D') \in \mathcal{S}] + \delta$. Therefore, $\mathcal{M}$ satisfies trajectory-level (ϵ, δ) -DP. $\square$

VI. EXPERIMENT

In this section, we compare various privacy protection methods and ST-ATT. We investigate the impact of our attention gate and other designs on ST-ATT. We also assess the proposed model with different parameters.

A. Datasets

We utilize two real-world datasets, Geolife and Porto:

Geolife [34]: This dataset includes 17,612 human trajectories gathered from 182 users in Beijing, China. It includes the users' GPS locations for a range of human outdoor activities.

Porto [35]: Assembled by 442 taxis in Porto, Portugal, this dataset encompasses over 1.7 million vehicle trajectories.

For experimental accuracy, we randomly select trajectories as our experimental subjects and remove the trajectories with fewer than ten sampling points. Consequently, we selected 9,000 trajectories from the Geolife dataset and 12,000 from the Porto dataset for our experiments.

B. Perturbation Models and Fairness of DP Comparison

'S' and 'ST': 'S' refers to adding noise satisfying DP only to spatial trajectory points. In contrast, 'ST' adds noise satisfying DP independently to spatial and temporal trajectory points without considering spatio-temporal correlations.

DPT [11] is a seminal paper that uses a hierarchical model to establish a reference frame for trajectories. This model encodes the conversion between the two by setting a grid cell tree with different resolutions through prefixes. Incorporating Laplace noise into the prefix tree distorts the transition probabilities to adhere to DP standards while partially preserving the original trajectory's motion patterns.

Frequency-based stochastic DP model [14] is a state-of-the-art work on perturbation trajectory point frequency for protecting trajectory privacy. This model considers the frequency of points on global trajectories and individual trajectories, raising the frequency of points low on global trajectories and lowering the frequency of points high on separate trajectories. This ensures no trajectory point possesses strong uniqueness.

SPRT [16] incorporates the public geographical structure of the target area into the synthesis of private trajectories, allowing for the capture of more precise movement patterns. This approach enables the creation of realistic trajectories that maintain aggregated-level statistics and individual-level mobility behaviors.

LDPTTrace [19] is an advanced paper that protects trajectory privacy from the perspective of local DP. This paper considers three key patterns inferred from user trajectories. It proposes a method for selecting an appropriate grid granularity for generating trajectories without compromising privacy, thereby achieving protection of trajectory privacy.

RetraSyn [20] protects trajectory privacy in the local DP setting. It focuses on real-time trajectory streams collected from location-aware devices, privately extracting mobility patterns via a global mobility model with dynamic updates, and then synthesizing high-utility trajectories on the fly through an adaptive privacy budget allocation strategy. RetraSyn allows for accurate analysis of streaming and historical trajectories without compromising privacy.

Fairness of DP Comparison: The baselines considered are developed under different privacy models, including pure DP, approximate DP, central DP, and local DP. We retain these baselines because they are representative and widely adopted in the

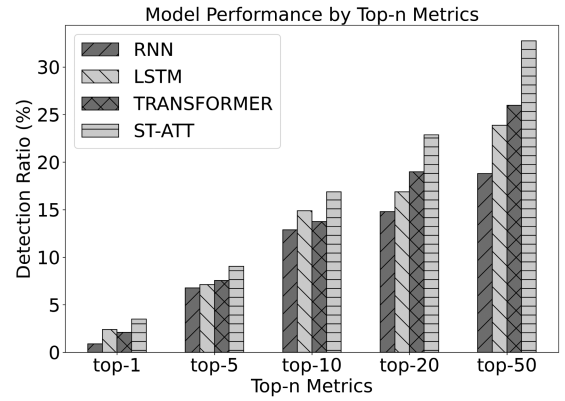


Fig. 3. Comparison between different classic detection models.

trajectory privacy literature. For baselines based on pure DP and Laplace noise, we also incorporate Gaussian-noise adaptation experiments, namely, replacing the original Laplace mechanism in these baselines with the Gaussian noise to better support a fair comparison. The results indicate that these adapted variants yield only limited utility gains (within 5% of the original results), remain overall inferior to our method, and reduce privacy protection performance by more than 5% relative to the original versions.

C. Evaluation Metrics (Privacy and Utility)

We use similarity search based on the Hausdorff distance as the query function in our DP approach to better evaluate the effectiveness of different noise addition methods.

A metric for evaluating privacy protection is $HR-k$, representing the top- k hit rate. We use $HR-1$, $HR-5$, $HR-10$, $HR-20$, and $HR-50$ as the primary evaluation metrics. The $HR-k$ metric indicates the probability of finding k trajectories in an unperturbed dataset that match the original trajectory of a perturbed trajectory. A higher $HR-k$ value suggests that more perturbed trajectories are identified, indicating that the privacy of the trajectory is not effectively protected. Another metric is MI [21], typically used to assess the dependency between original and anonymized datasets. Lower MI indicates more effective data protection.

We also use two metrics, Diameter Distribution (DE) and TE from [11], to measure the utility of the data. To calculate the DE and TE for temporal trajectories, we divide the time into 25 equal intervals per day. $E_{s-diameter}$ represents the DE of spatial trajectories, and is evaluated using [11, Eq. (14)]. $E_{t-diameter}$ represents the DE of temporal trajectories and can be evaluated in the same way as the DE of spatial trajectories. Similarly, E_{s-trip} represents the TE of spatial trajectories and is calculated [11, Eq. (15)]. E_{t-trip} represents the TE of temporal trajectories and can also be evaluated using [11, Eq. (15)]. We use $E_{diameter}$ and E_{trip} to represent the DE and TE of spatio-temporal trajectories, respectively. To better express the relationship between spatial and temporal dimensions, we define $E_{diameter} = \lambda \cdot E_{s-diameter} + (1 - \lambda) \cdot E_{t-diameter}$ and $E_{trip} =$

$\lambda \cdot E_{s\text{-trip}} + (1 - \lambda) \cdot E_{t\text{-trip}}$, where λ adjusts the relevant importance of the spatial and temporal dimensions. By default, we set $\lambda = 0.5$, unless otherwise specified.

Global and semantic utility metrics: We further adopt four global- and semantic-level metrics to evaluate the utility of synthetic trajectories: We discretize the spatial domain into a uniform $N \times N$ grid. Given the bounding box of all points, both axes are split into N equal segments, yielding N^2 square cells. Following the guideline of LDPTrace [19], we choose N by trading off perturbation and range-boundary errors. On our datasets, this gives $N = 12$, which is used for all methods. Discrete timestamps are grouped into windows of length φ . In line with RetraSyn [20], we set $\varphi = 20$ timestamps when computing utility metrics.

Density Error (DenE): At each timestamp we count visits of T^{st} and $T^{\text{st}'}$ in every grid cell and obtain two histograms. DenE is the Jensen–Shannon divergence between these histograms, averaged over all timestamps.

Range Query Error (RQE) / Query Error: A range query is defined by a random spatial window and a time interval of length φ . For a dataset D , $Q_i(D)$ returns the number of points within this space–time region. RQE is the mean relative error between $Q_i(T^{\text{st}})$ and $Q_i(T^{\text{st}'})$ over 100 random queries.

Hotspot Query Error (HQE): We rank grid cells by visit frequency and take the top- k hotspots in T^{st} and $T^{\text{st}'}$, denoted H_{st} and $H_{st'}$. The agreement of the two rankings is measured by NDCG@ k , and we define HQE = $1 - \text{NDCG}@k$. A smaller HQE indicates that spatial hotspots are better preserved. We use $k = 10$ by default.

Pattern F1: A semantic mobility pattern P is modeled as an ordered sequence of consecutive cells. Within a time window of length φ , we extract the top- N most frequent patterns from T^{st} (reference set) and from $T^{\text{st}'}$ (retrieved set), compute precision and recall of synthetic patterns w.r.t. the original ones, and take their harmonic mean as the F1 score for that window. The reported Pattern F1 is the average over 50 random windows.

D. Experiment Settings

We implement ST-ATT and other non-open-source models, including SDD and Frequency DP in the Python PyTorch framework. 20% of the trajectory data is used for training and 80% for testing. The optimizer is Adam, with the learning rate 5×10^{-3} and the dimension of the feature vectors ($d = 128$). Before being fed into the models, all trajectories are normalized and padded to the same length. The spatial domain is discretized into a uniform 12×12 grid. The temporal window parameter is set to 20. For model training, we use both full-trajectory and sub-trajectory supervision, with the loss weight $\alpha = 0.8$. For privacy protection, the proposed mechanism perturbs both the spatial and temporal components under the overall privacy budget (ϵ, δ) , where $\delta = 10^{-5}$. All compared methods are evaluated on the same datasets under the same task setting and evaluation protocol to ensure fairness. The evaluation metrics include Top- k hit rate, Mutual Information (MI), Trip Error (TE), Hausdorff Distance, and other utility measures. All experiments are repeated five times, and the averaged results are reported.

The experiments are conducted on a machine equipped with an Intel Xeon E-2288 G 8-core CPU and an NVIDIA Quadro RTX A5000 GPU.

E. Performance Evaluation

Using the ST-ATT model under Hausdorff distance-based similarity learning, we compare the extent of trajectory data protection offered by different noise addition methods. This approach demonstrates the effectiveness of various noise addition methods in protecting trajectory data. For the Geolife dataset, we select 9,000 trajectories, of which 7,200 are used as test data. We randomly choose 12,000 trajectories for the Porto dataset, with 9,600 for testing.¹

All comparisons are conducted with each trajectory assigned the same privacy budget. The privacy budget for an entire trajectory is set to $\epsilon = 1$ to ensure fair comparisons. As shown in Tables II and III, ‘S’ adds DP noise only to the spatial trajectory, with each spatial trajectory point receiving the same privacy budget. No noise is added to the temporal trajectory before being published. ‘ST’ indicates that DP noise is added to both spatial and temporal trajectory points, with a privacy budget of 0.5 allocated to each of the temporal and spatial trajectories. The same noise is added to each spatio-temporal trajectory point. ‘ST-Cor’ represents Algorithm 1. Other comparison methods also generate synthetic perturbed trajectories under the same trajectory and privacy budget. We then test these using the ST-ATT detection model.

Table II shows global attention, while Table III represents local attention, meaning only a subset of related temporal trajectory points participate in the attention computation. Bold-face indicates the best experimental results under the same metrics. The percentages of the HR-1, HR-5, HR-10, HR-20, and HR-50 metrics for the ‘ST-Cor’ noise addition method are significantly lower than those of other methods, indicating that the probability of perturbed trajectories generated by ‘ST-Cor’ being identified by ST-ATT is considerably lower than that of perturbed trajectories generated by existing methods. This demonstrates the rationality of adding noise according to the quantitative spatio-temporal privacy budget relationship in DP noise. Our noise addition scheme is the most effective in protecting trajectory privacy.

In addition to the HR- k metric, we use the MI index to measure the degree of privacy protection of the trajectory, and we employ the metrics DE and TE to evaluate its utility. As shown in Table IV, we conduct evaluations of the MI, DE, and TE metrics on the Porto dataset. The results show that the MI index of ‘ST-Cor’ is significantly lower than that of existing noise protection models. This indicates that ‘ST-Cor’ is considerably better at privacy protection than other models, which is consistent with the results of HR- k . This suggests that a reasonable allocation of the privacy budget can effectively generate noise trajectories with better privacy. Regarding the DE metric, ‘ST-Cor’ outperforms most privacy protection methods,

¹Subsequent experiments demonstrate that a dataset of around 2,000 trajectories is sufficient to train an effective model. Using more data for testing here can make the experimental results more accurate and reduce randomness.

TABLE III
PERFORMANCE OF LOCAL ATTENTION (IN %)

	Geolife					Porto				
	HR@1	HR@5	HR@10	HR@20	HR@50	HR@1	HR@5	HR@10	HR@20	HR@50
S	6.24	12.78	19.87	25.11	36.98	11.34	24.56	28.88	35.77	58.56
ST	2.78	5.08	8.26	12.25	17.86	2.36	9.97	18.25	27.86	39.56
F [14]	6.04	11.67	16.53	21.88	26.97	18.75	38.99	55.87	68.98	79.56
DPT [11]	3.88	8.45	12.66	16.83	24.11	21.6	42.47	53.11	67.88	78.76
SPRT [16]	2.84	5.12	9.25	11.28	16.55	13.68	30.12	40.35	52.65	60.54
LDPTTrace [19]	2.56	6.09	9.55	12.43	16.62	3.66	7.22	13.32	17.45	23.12
RetraSyn [20]	2.28	5.54	8.40	11.19	15.29	3.18	6.71	11.99	15.36	21.04
ST-Cor	0.38	0.89	1.96	3.01	4.65	0.13	0.42	0.98	1.87	4.12

TABLE IV
COMPARISON OF DE, TE, AND MI ON THE PORTO DATASET

	DE	TE	MI
S	0.365	0.423	0.109
ST	0.189	0.401	0.068
F	0.018	0.331	0.095
DPT	0.297	0.375	0.097
SPRT	0.285	0.386	0.076
LDPTTrace	0.276	0.329	0.072
RetraSyn	0.243	0.296	0.062
ST-COR	0.125	0.296	0.036

though it is not as good as the ‘F’ method. This is because the ‘F’ method only changes the frequency of trajectory points without altering their longitude and latitude, which makes the DE metric particularly favorable for this privacy protection method.

Changing the trajectory point frequency is susceptible to linkage attacks, resulting in poor privacy protection. The MI and top metrics also show that merely altering the frequency of trajectory points is not reasonable. However, regarding the TE metric, the TE value of ‘ST-Cor’ is lower than that of other privacy protection methods, indicating that the ‘ST-Cor’ privacy protection method significantly outperforms other privacy protection methods. ST-Cor performs well in both TE and DE metrics, demonstrating that our privacy protection method provides robust privacy protection and high utility.

Global and semantic utility: We consider three baselines on the Porto dataset: ST, LDPTTrace [19], and RetraSyn [20] (Table V). We focus on LDPTTrace and RetraSyn because they represent the most recent and competitive locally differentially private trajectory synthesis methods. RetraSyn is designed for spatio-temporal trajectories, instead of static spatial data. ST serves as the DP-based backbone without correlation-aware budget allocation; LDPTTrace and RetraSyn are state-of-the-art LDP synthesis schemes. Boldface highlights the best experimental result under each metric and privacy budget.

Across all privacy budgets $\epsilon \in \{0.5, 1.0, 1.5, 2.0\}$, ST-Cor consistently achieves the lowest Density Error (DenE), Range Query Error (RQE), and Hotspot Query Error (HQE), and the highest Pattern F1 score. For example, when $\epsilon = 1.0$, ST-Cor reduces DenE from 0.203 (ST) and 0.163 (LDPTTrace) to 0.131, and lowers HQE from 0.284 (ST) and 0.239 (LDPTTrace) to 0.182, while improving F1 from 0.436 (ST) and 0.495 (LDPTTrace) to 0.551. ST-Cor not only preserves the global spatio-temporal density and range-query answers more accurately, but also better maintains hotspot structures and frequent mobility patterns than existing synthesis methods. The consistent gains

over ST demonstrate that modeling spatio-temporal correlations and allocating the privacy budget accordingly is an effective and well-justified design choice for trajectory synthesis under DP.

F. Detection Model Comparison

We evaluate the proposed noise addition method in comparison with several standard sequence processing models (i.e., RNN, LSTM, Transformer). RNN continuously updates its hidden state to preserve sequence information, making it suitable for handling variable-length sequences with long-term dependencies in trajectory data. LSTM controls the flow of information through gating units, so it is effective in processing sequences that require long-term memory. The Transformer, a model based on self-attention mechanisms, is suited for parallel processing of long sequences and can capture relationships between different positions within the sequence. These three models are classical in addressing sequence problems and adept at capturing the internal correlations within trajectories. For fair comparisons, we add noise to spatio-temporal trajectories and compare the accuracy of these models in matching the true trajectories. Fig. 3 shows our model outperforms these traditional detection models in all metrics.

We further select several models with linear complexity that are effective in noise detection (NeuTraj, T3S, and ST-T2SV). NeuTraj uses a metric learning approach to trajectory similarity learning, adding a newly proposed memory unit called SAM to the LSTM for storing and retrieving historical information of processed trajectories. T3S, the most efficient model, proposes a neural network combining LSTM and self-attention to learn spatial and structural information of trajectories based on networks. ST-T2SV is the state-of-the-art method for trajectory similarity learning, leveraging sub-trajectory information within trajectories to enhance model judgment with additional auxiliary information. ST-ATT inherits the approach to incorporating sub-trajectory information from ST-T2SV and modifies the LSTM by adding an attention gate to better learn spatio-temporal correlations and internal connections within trajectories. Fig. 4 shows that our model achieves the highest top- k identification for perturbed trajectories.

G. Efficiency

We evaluate the time efficiency of both traditional matching algorithms and learning-based algorithms. The total time is categorized into three components: training, inference, and computation. Training time denotes the duration required for a learning-based model to complete one training cycle. Inference

TABLE V
GLOBAL AND SEMANTIC UTILITY METRICS UNDER DIFFERENT PRIVACY BUDGETS ϵ ON THE PORTO DATASET

Method	$\epsilon = 0.5$				$\epsilon = 1.0$				$\epsilon = 1.5$				$\epsilon = 2.0$			
	DenE	RQE	HQE	F1	DenE	RQE	HQE	F1	DenE	RQE	HQE	F1	DenE	RQE	HQE	F1
ST	0.214	0.365	0.302	0.412	0.203	0.348	0.284	0.436	0.196	0.332	0.271	0.452	0.188	0.318	0.259	0.467
LDPTTrace	0.176	0.308	0.253	0.471	0.163	0.293	0.239	0.495	0.151	0.276	0.224	0.512	0.143	0.263	0.213	0.532
RetraSyn	0.168	0.296	0.241	0.486	0.158	0.281	0.228	0.507	0.147	0.265	0.213	0.525	0.139	0.254	0.203	0.541
ST-Cor (Ours)	0.143	0.259	0.199	0.521	0.131	0.241	0.182	0.551	0.122	0.226	0.168	0.572	0.115	0.214	0.157	0.593

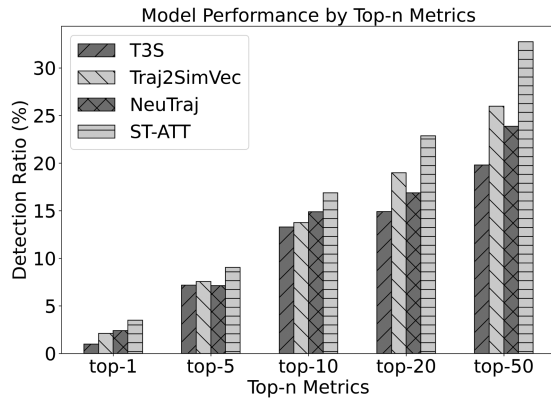


Fig. 4. Comparison between T3S, NeuTraj, Traj2SimVec, and ST-ATT.

TABLE VI
EFFICIENCY STUDY OF EXACT DISTANCE METRICS AND LEARNING-BASED METHODS

Dataset	Method	Time Cost (s)		
		Training	Inference	Computation
Porto	Hausdorff	/	/	1680
	NeuTraj	104.58	0.00059	0.0002
	T3S	55.36	0.00023	0.0002
	ST-T2SV	220.89	0.00032	0.0002
	ST-ATT	20.68	0.00021	0.0002

time represents the average time the model takes to encode trajectories during the testing phase. Computation time refers to the time required to compute the similarity between two vectors using the learning model. The time consumption of the methods is detailed in Table VI. For this evaluation, we randomly select 2,000 trajectories from the Porto dataset as the test set. The detection time of the traditional Hausdorff algorithm is 1,680 seconds. The learn-based model only calculates the Euclidean distance between the 128-dimensional vectors, so it only takes 0.2 ms.

There is no significant difference in training time among the models. The time for each epoch of ST-ATT, NeuTraj, T3S, and ST-T2SV models to process 2,000 tracks is 20.68, 104.58, 55.36, and 220.89 seconds, respectively. Regarding coding time, ST-ATT only takes 0.21 ms to encode a trajectory. The slowest method, NeuTraj, takes 0.59 ms due to the time-consuming operations of memory read/write. Overall, all learning-based methods are very efficient.

We compare the time cost of similarity computation for different trajectory sizes. We randomly select 1 k, 5 k, 10 k, and 200 k Porto trajectory data. As shown in Table VII, Hausdorff

TABLE VII
TIME COST FOR COMPUTING SIMILARITY WITH VARIOUS SIZES ON PORTO DATASET

Method	1k	5k	10k	200k
Hausdorff	408.1s	9956.8s	40034.9s	N/A
T3S	3.9s	15.9s	32.98s	510.93s
ST-ATT	3.6s	15.58s	30.38s	498.09s

represents the traditional algorithm, ST-ATT is our method, and T3S is currently the most efficient method. It is observed that learning-based algorithms are over a hundred times faster than traditional algorithms. Our algorithm is nearly as efficient as T3S, and slightly faster.

H. Ablation Study

We conduct an ablation study on this model. The attention gate model is removed from the ST-ATT, creating a variant called ‘ST-ATT w/o A’. We eliminate the attention gate from ST-ATT, while the preprocessing of spatio-temporal trajectories and other network structures remains the same as in ST-ATT. We employ the ‘ST-ATT w/o A’ network to perform similarity detection tasks on two datasets. The results, as shown in Fig. 5(a), indicate that after removing the attention mechanism, in all scenarios, ‘ST-ATT w/o A’ performs worse on all evaluation metrics than the ST-ATT model.

We conduct an ablation study on this model to demonstrate the importance of our proposed temporal preprocessing model. The temporal processing model is removed from ST-ATT, naming this variant ST-ATT w/o T. We eliminate the cyclic learnable function for the time being within ST-ATT, directly using time series data as input to the model. The rest of the network structure remains consistent with ST-ATT. We also employ the ST-ATT w/o T network to perform similarity detection tasks on two datasets. Fig. 5(b) indicates that after removing the temporal processing model, in all scenarios, ST-ATT w/o T performs worse on all evaluation metrics than ST-ATT. Fig. 5(c) shows the impact of different loss functions on various metrics in ST-ATT, where we compare the MSE with RMSE and Cosine Similarity Loss functions. The MSE loss function calculates the squared difference between the predicted and the true values. RMSE is the square root of MSE, making it more sensitive to outliers.

I. Impact of Parameters

We conduct experiments on the Porto dataset to assess the impact of various parameters on the model. We adjust the dimensionality d of the spatio-temporal trajectory high-dimensional vectors in the ST-ATT model from 32 to 512 dimensions. A large

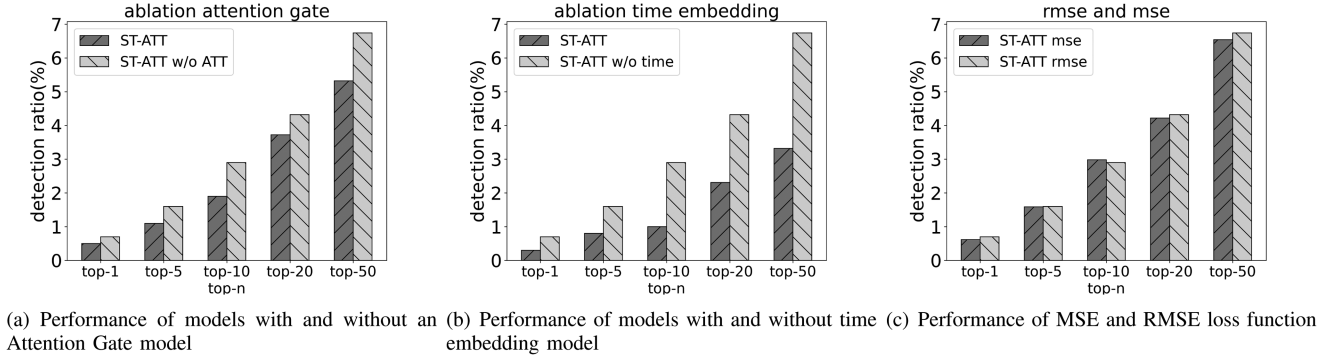


Fig. 5. The ablation study of Attention gate, time embedding, and loss function of the proposed ST-ATT on the Porto dataset.

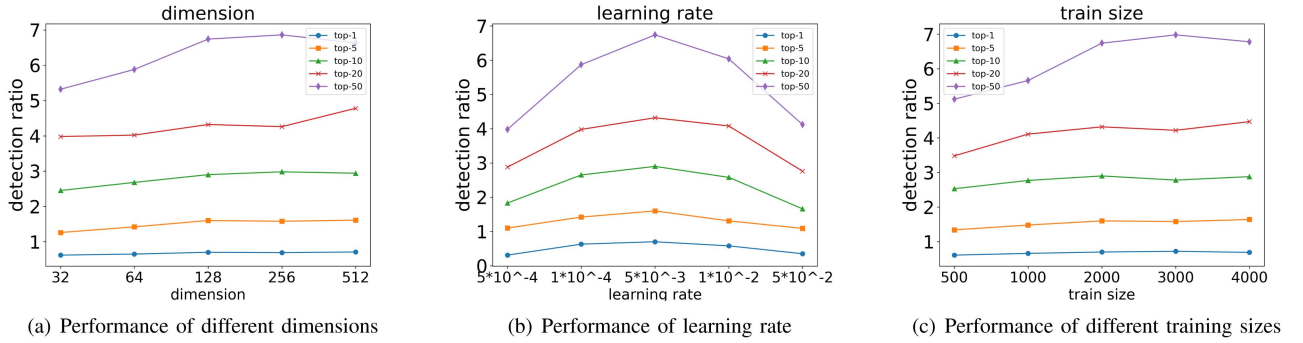


Fig. 6. The performance of ST-ATT versus the dimension d , learning rate lr , and training size.

d increases the time and space costs of the learning parameters, potentially decreasing model efficiency; a small d limits the model's expressiveness. Fig. 6(a) demonstrates that the optimal performance is achieved with a representation dimension of 128. We also conduct experiments with learning-based models to assess the space costs associated with different dimensional settings. During training, storing spatio-temporal trajectory data for 2,000 trajectories requires 2 MB. Since the required storage space scales linearly with the number of trajectories, it is crucial to select an appropriate number of trajectories and representation dimensions.

We change the learning rate (lr) from 1×10^{-4} to 1×10^{-2} . A high learning rate, e.g., 1×10^{-2} , leads to suboptimal performance in the ST-ATT model. The reason is that excessive oscillations in the loss function hinder model convergence. Conversely, a low learning rate requires intensive learning from the model. Fig. 6(b) indicates 5×10^{-3} is the optimal learning rate for ST-ATT on the Porto dataset.

We adjust the training sample size from 500 to 4,000 to assess the impact of training sample size on the approximation accuracy of ST-ATT. We select 2,000 trajectories as the training set. Fig. 6(c) plots the performance of our model across various training sizes, including 2,000. ST-ATT's performance stabilizes when the number of training samples exceeds 2,000, suggesting that a small number of training trajectories can effectively train our model to recognize original trajectories. Even with small training sizes, e.g., 500 or 1,000, ST-ATT performs well, demonstrating its robustness and capability to learn effectively from small datasets.

VII. CONCLUSION

This paper quantifies the relationship between the privacy budgets of spatial and temporal points in trajectories by accounting for their inherent correlations, and develops a new spatio-temporal DP mechanism to enhance the protection of trajectory points. Specifically, we demonstrate that the privacy budget of a spatial trajectory point depends solely on the privacy budget of its associated temporal trajectory point and the probability of violating DP for that point. To validate the proposed mechanism, we develop a novel trajectory-matching model, ST-ATT, which incorporates spatio-temporal correlation learning and time embedding into the LSTM framework. Experiments confirm that the proposed spatio-temporal DP mechanism outperforms existing methods for trajectory privacy, demonstrating robust defense against linkage attacks, including those executed by advanced models, e.g., ST-ATT.

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